

Tying the (Data) Knot: Love, Life & Likes

Predicting Dating App Match Outcomes Using Machine Learning

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Problem & Context

Background

Dating apps generate behavioral signals from user interactions. This project studies whether such signals (alongside user attributes) can predict match outcomes in a synthetic setting.

Problem Statement

Predict match_outcome (10 classes) from 24 features on 50,000 synthetic users.

TASK SUMMARY

TARGET

match_outcome

CLASSES

10 Classes

INPUTS

24 Features

DATASET

50,000 Users

TASK TYPE

Supervised Multi-class Classification

OBJECTIVES

Pipeline

Design and document a complete, reproducible machine-learning pipeline for predicting dating-app match outcomes.

Comparison

Train, tune, and rigorously compare at least five machine-learning models on the same processed data.

Benchmarking

Benchmark the best model against an automated machine-learning (AutoML) baseline to validate the ceiling of achievable performance.

Interpretation

Critically interpret the results to draw an honest, evidence-based conclusion about whether these behavioral signals predict match outcomes.

Methodology and Model Explanation

01

Project Design

Framed as supervised multi-class classification predicting match outcome, combining multiple relationship phenomena into a single label — across 10 output classes.

02

Data Collection and Acquisition

The provided synthetic dataset is an extended version of the Kaggle "Dating App Behavior Dataset" containing 50,000 records and 25 columns. After loading and removing duplicates, the dataset retained its full 50,000 records. The features span four broad groups:

Feature Group	Example Features
Demographics	gender, sexual_orientation, age, height_cm, weight_kg, body_type, zodiac_sign
Socio-economic	income_bracket, education_level, location_type
App usage/behavior	app_usage_time_min, swipe_right_ratio, likes_received, mutual_matches, message_sent_count, emoji_usage_rate, last_active_hour, swipe_time_of_day, profile_pics_count, bio_length
Stated preference	interest_tags, relationship_intent
Target	match_outcome (10 classes)

03

Data Pre-processing

Q Redundant-label removal ×

Two redundant textual columns (app_usage_time_label and swipe_right_label) were dropped to prevent multicollinearity.

Q Outlier capping ×

Four heavily skewed numeric features were clipped to their 1st and 99th percentiles to prevent extreme values from distorting models.

Q Cyclical encoding ×

The last_active_hour feature was encoded into two continuous features using sine and cosine transforms to reflect its circular nature.

Q Ordinal encoding ×

education_level and income_bracket were encoded with an explicit hierarchy reflecting their natural ranking.

Q One-hot encoding ×

Seven unordered categorical features were converted with a OneHotEncoder.

Q Scaling ×

All remaining numeric features were scaled with a RobustScaler, which is resistant to outliers.

04 Feature Selection and Extraction

114 Features



Interest Binarisation
expanded interest_tags into binary columns



Variance Filtering
removed near-constant columns



53 Features



Mutual Information
SelectKBest - top 30 scoring features



30 Features



PCA Check
25 components needed for 95% variance -
minimal gain



30 Named Features Retained
interpretability preserved

A Critical Early Warning:
Mutual-information scores were near zero — even the strongest feature (age) scored just 0.0077, indicating almost no statistical dependency with the target.

04 Model Selection

Logistic Regression

A linear baseline that models class probabilities directly.

Random Forest

A bagging ensemble of decision trees that reduces variance through bootstrap aggregation.

Extra Trees

A more randomised variant of Random Forest that splits on random thresholds.

Gradient Boosting

A sequential boosting method that fits each tree to residual errors.

XGBoost

An optimised, regularised gradient-boosting implementation known for strong tabular performance.

LightGBM

A fast, leaf-wise gradient-boosting framework highly efficient on large datasets.

05 Model Training and Hyperparameter Tuning

- Tuning method: RandomizedSearchCV using 3-fold cross-validation
- Scoring metric: Macro-F1 (weights every class equally)
- Each tuned model evaluated on held-out 30% test set

06 Model Evaluation

- Evaluated on: Macro-F1, Accuracy, Training time
- Confusion matrix generated for best model (LightGBM)
- Best model benchmarked against auto-sklearn

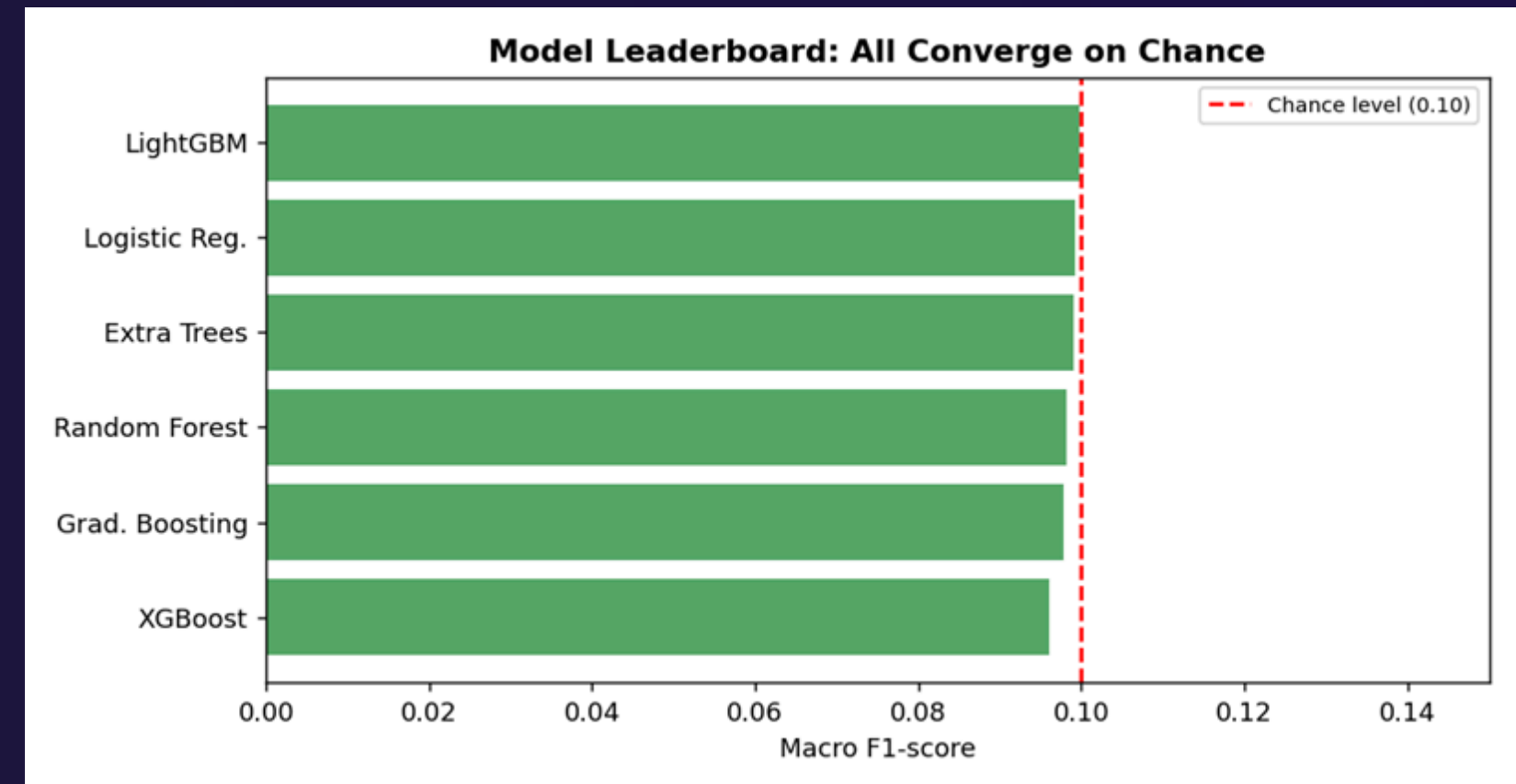
Results and Visualization

01

Model Leaderboard

After tuning, all six models were evaluated on the held-out **test set**. The results show that **all models cluster tightly around 0.10** in both F1 and accuracy:

Model	Macro F1	Accuracy	Time (s)
LightGBM	0.0997	0.0998	172.93
Logistic Regression	0.0993	0.1025	33.12
Extra Trees	0.0991	0.0991	135.66
Random Forest	0.0982	0.0987	264.68
Gradient Boosting	0.0979	0.0991	1096.95
XGBoost	0.0961	0.0961	121.27

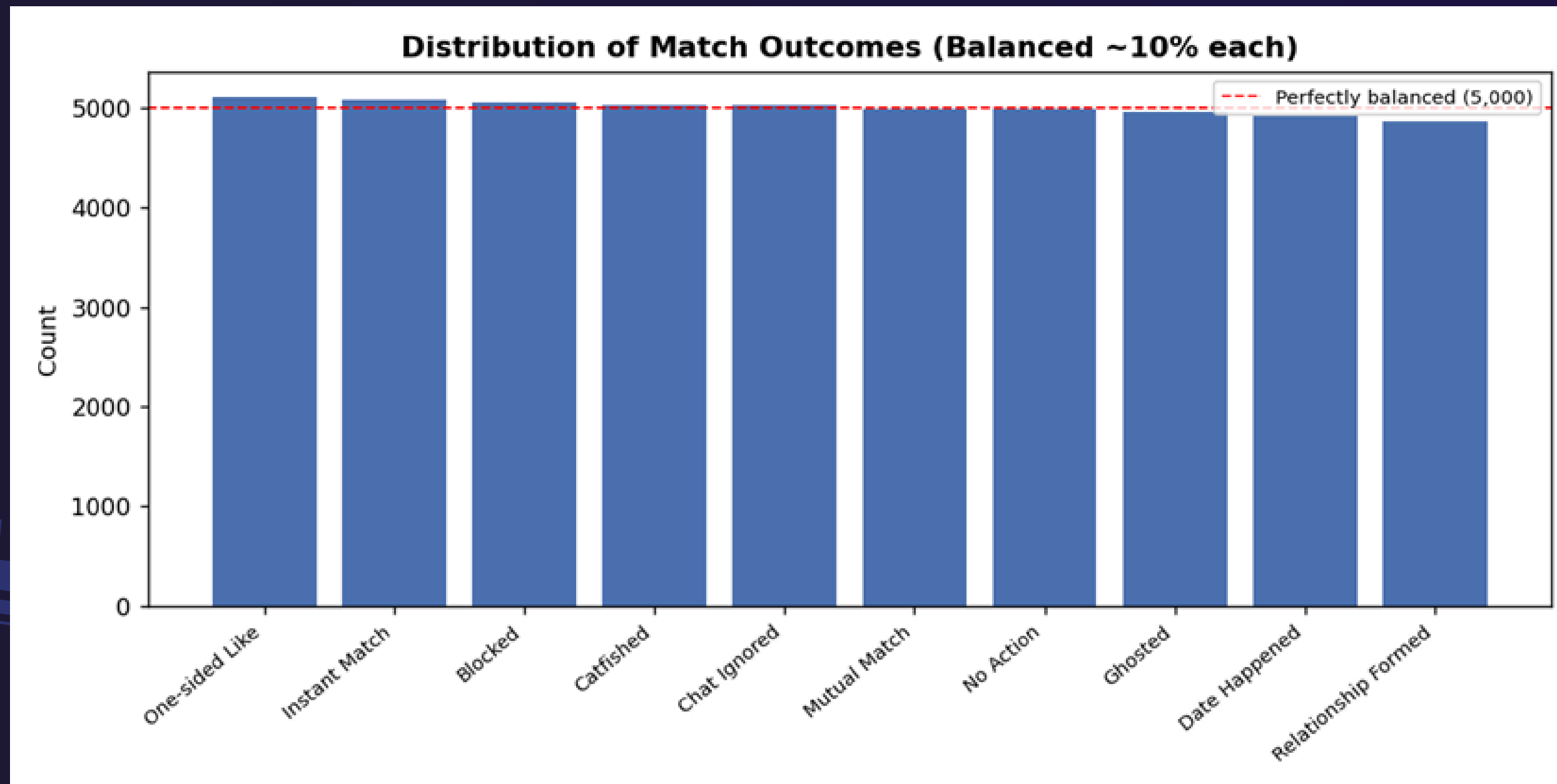


The **best model**, LightGBM, achieves **0.0998** accuracy, statistically indistinguishable from the 10% chance level. The spread between the best and worst model is **less than four-tenths of one percent** — the signature of a problem with **no signal to learn**.

Results and Visualization

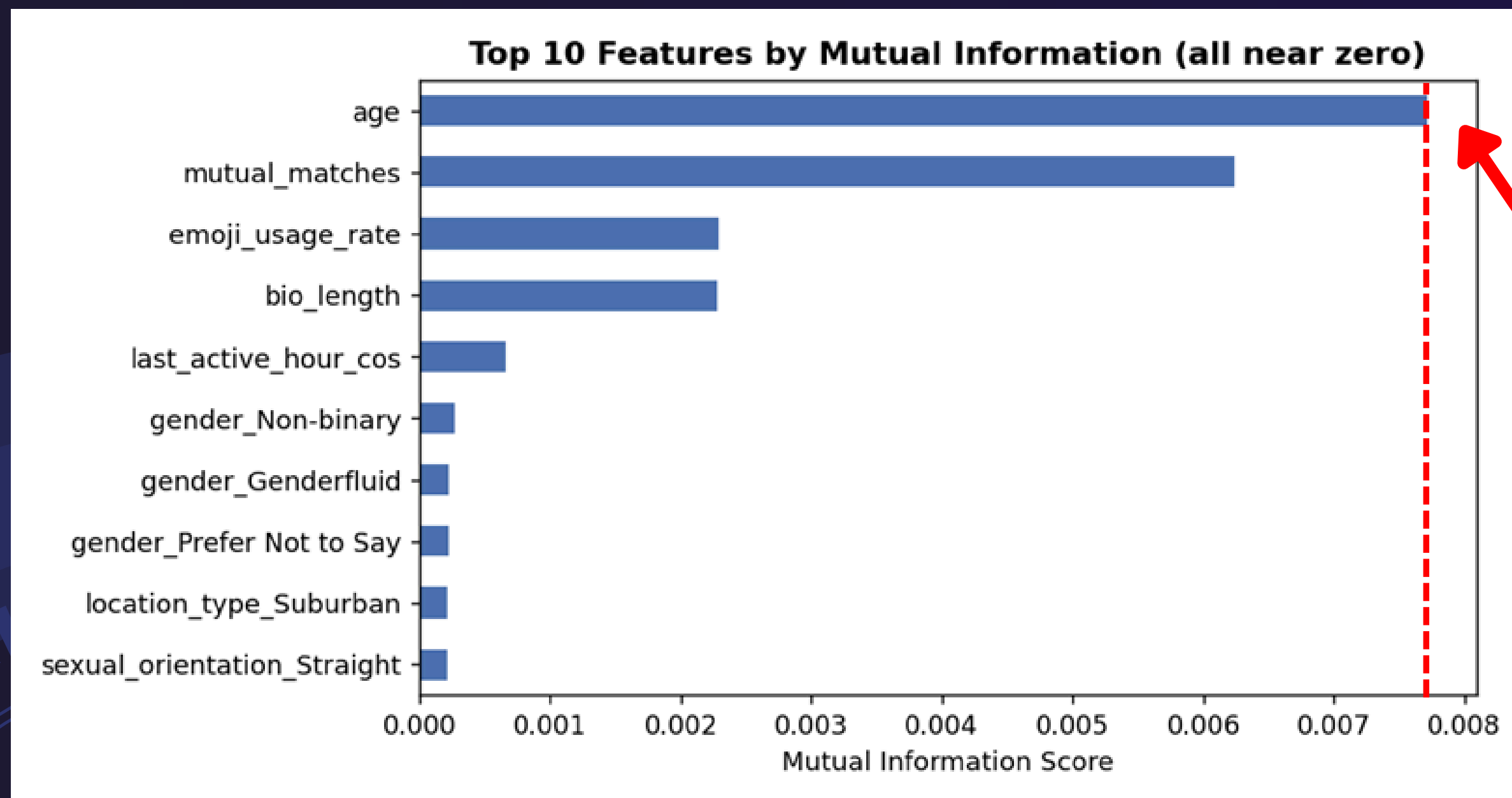
02 Class Distribution

Ten perfectly balanced classes (~10% each) where no single outcome dominates — meaning that 10% accuracy is the random chance baseline.



03 Feature Informativeness

The mutual-information analysis revealed that no feature carries meaningful predictive information about the target.

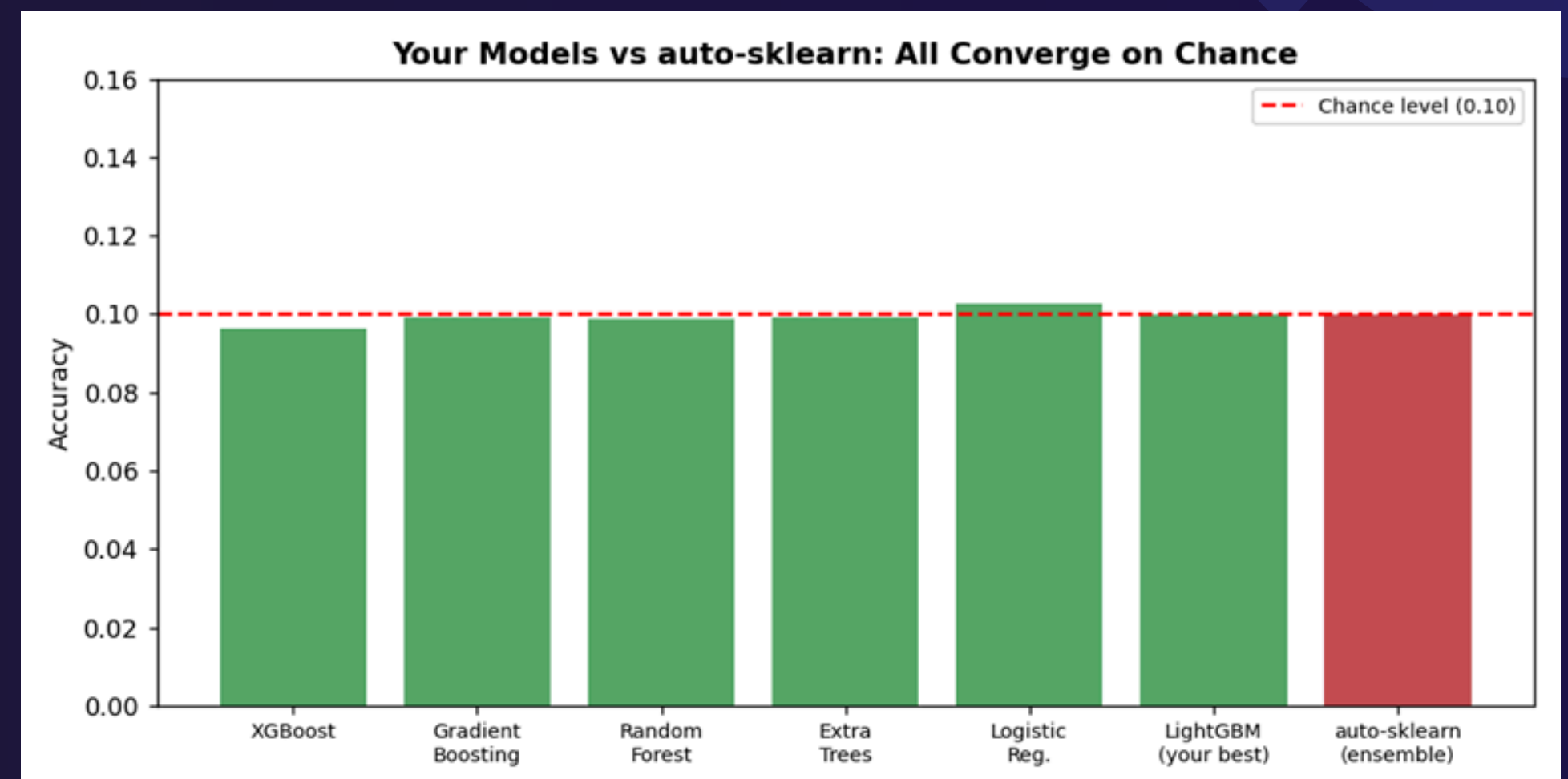


Even the **strongest** feature (**age**) scored just **0.008**, with most others near zero, offering no **real signal**.

04 Comparison with auto-sklearn

The pipeline was benchmarked against auto-sklearn, an AutoML framework searching across algorithms, hyperparameters, and preprocessing steps to build an optimised ensemble:

Approach	Accuracy	Macro F1
Our best model (LightGBM)	0.0998	0.0997
auto-sklearn ensemble (LDA + 2× Gradient Boosting)	0.0999	0.0997
Chance level (balanced 10-class)	0.1000	—



Result:

Auto-sklearn and LightGBM (the best model) both hit 10% chance level, confirming that the limitation is in the data, not the model.

Insights and Interpretation

The Central Finding: A Dataset Without Signal

Three independent lines of evidence confirm no learnable relationship between features and match outcome:

Mutual-information scores are uniformly near zero.

Six diverse, tuned models all perform at the 10% chance level.

Auto-sklearn given free rein to search and ensemble, confirms the same ceiling.

When linear models, bagging ensembles, boosting ensembles, and AutoML all arrive at identical chance-level performance, the only consistent explanation is that the target was generated independently of the features.

Why This Happens

The outcome labels appear randomly assigned — when every model and AutoML converge at chance level, no real pattern was ever encoded in the data.

What the Methodology Demonstrates

A low result is not a methodological failure. By avoiding data leakage and training-set evaluation, our pipeline established the absence of signal rigorously — a trustworthy negative result is more valuable than a false positive.

Practical and Ethical Reflection

Predicting ghosting or match success from gender or body type risks teaching the algorithm to discriminate, amplifying real-world bias. Not every outcome should be predicted — reporting that honestly is responsible machine learning.

Conclusion

- A complete **ML pipeline** was built to **predict dating-app match outcomes from user behaviour** — covering data cleaning, feature engineering, and six tuned classifiers.
- The central finding: The dataset contains **no learnable signal** — all six models and auto-sklearn converged at **10% accuracy**, identical to random guessing, confirming the limitation is intrinsic to the data.

Limitations



Synthetic Data

Target labels appear randomly assigned — no real behavioural signal present.



Metric Ceiling

Accuracy and F1 can only confirm absence of learning, not rank models.



Compute Budget

Longer AutoML search would not change a chance-level conclusion.

- An insightful summary: **Love, life, and likes simply cannot be predicted.**

Thank You



Tying the (Data) Knot: Love, Life & Likes

Links

5-minute Video Presentation

<https://drive.google.com/file/d/1DBArGC434yJKsam-wicNxHLxKPiMFBDd/view?usp=sharing>

Google Colab

https://colab.research.google.com/drive/1KrbUu9vnd13_hfug7iCcdWaqGzCGCW5v?usp=sharing